Rating Prediction of Football Players using Machine Learning

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ABSTRACT

This research Analysis the prediction of football player ratings through the application of diverse machine learning algorithms. Rating systems for sports teams have gathered considerable attention in academic research. The approach used by the authors of this paper serves as an effort to streamline scouts and performance analytics. Leveraging linear regression, decision tree regressor, random forest regressor, gradient boosting regressor, support vector regressor, voting regressor, ridge regression, lasso regression, k-nearest neighbours' regression, Huber regression and elastic-net regression. The Analysis explores the efficiency of each algorithm and concludes that Support Vector Regressor algorithm performs the best with 91.84% accuracy on the testing data followed by the Gradient Boosting Regressor with 90.78%, Voting Regressor with 91.68% and Random Forest Regressor with 88.89%. Apart from them the K-Nearest Neighbours Regression Algorithm highly overfits the model with 100% accuracy on the training set and 70.71%. The conclusions drawn underscore the critical importance of judiciously selecting algorithms tailored to the specific characteristics of the dataset for precise and reliable player rating predictions.

CCS CONCEPTS

• **Computing methodologies** → Machine learning; Learning paradigms; Supervised learning; Supervised learning by regression.

KEYWORDS

Ratings, Machine Learning, Algorithms, Regression Algorithms and Football Players

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Figure 1: Player Card

1 INTRODUCTION

The increasing demand for the football player rankings and data driven performance analysis has become increasingly popular within the football industry. As the task of talent selection from the extensive pool of soccer players becomes more and more intimidating, the integration of data-driven performance scores emerges as an indispensable tool. The scores generated helps in identification of a more defined and accurate subgroup of top-performing players, ranging from age and performance characteristics to positional roles and evolving trends. This approach not only streamlines the efforts of scouts and performance analysts but also leads to substantial time and cost savings while simultaneously broadening the scope of scouting operations and offering promising career prospects for talented players. Figure 1. Represents the Player performance card.

The development of rating system for sports teams has got a significant attention in research purposes. The development of rating systems for sports teams has attracted. With respect to above mentioned factors, the idea of building a machine learning model came into the picture. This research aims to predict the ratings of football players using a diverse array of machine learning algorithms. These algorithms include Decision Tree Regressor, Random Forest Regressor, Linear Regression, Gradient Boosting, Support Vector Regressor, Logistic Regression, Voting Regressor, Huber Regressor, KNN, ElasticNet Regressor, Ridge Regression, and Lasso Regression.

The paper is summarized as follows: Section 2 described previously published research in the area, Section 3 summarizes the methodologies used for EV Range prediction, Section 4 summarizes the result obtained from various models and at lastly draws the conclusion and future enhancement.

2 LITERATURE AND SURVEY

In the paper [1], the authors have elucidated the correlation between player ratings, employing the methodologies of absolute dispute resolution through the utilization of gradient descent and stochastic gradient descent techniques.

The authors in [2] developed a data-driven framework known as "PlayeRank," which offers a role-aware analysis of the evolutionary trajectory of soccer player performance. This framework involves a comprehensive comparison of soccer player performance by leveraging established algorithms.

The authors introduced a classification methodology aimed at selecting the most suitable player from a pool of talented candidates to constitute a soccer team [3]. Their approach encompasses the application of various machine learning algorithms, including supervised learning techniques, decision trees, and random forests.

The authors successfully implemented a predictive model designed to forecast player performance [4]. In this predictive analysis, they harnessed the power of multiple machine learning algorithms, namely Decision Trees, Naive Bayes, Random Forest, and Multiclass Support Vector Machine (SVM).

In [5] the authors introduced a comprehensive system designed for the ranking of players and the prediction of tournament outcomes. Their approach leverages network analysis techniques and methodologies to enhance the accuracy of player ratings.

The authors in [6] devised a predictive model for forecasting football match results through the utilization of machine learning algorithms. This model serves as a valuable tool in evaluating team performance.

The authors in [7] introduced a sophisticated multi-level regression methodology designed for the estimation of players' market values. To facilitate this analysis, they meticulously compiled a dataset that incorporates diverse attributes, encompassing player characteristics like age, position, and nationality, as well as variables related to player performance and popularity.

The authors conducted an in-depth investigation to assess the influence of diverse factors on the valuation of forward players, with the aim of identifying the most pivotal contributors to player value [8].

The authors in [9] introduced a novel method for the estimation of transfer fees for football players, relying on the consideration of five essential talent variables: age, precision, success, assertion, and flexibility.

The authors introduced a comprehensive methodology for the estimation of the market values of 12,858 players [10]. This estimation is founded on the analysis of players' performance data. Their models were meticulously crafted, incorporating a total of 45 predictor variables, allowing for a thorough and data-driven approach to estimating player market values.

3 METHODOLOGY

3.1 Dataset

To build the machine learning model our study includes the database of 18000+ football players, among which there are a total of 35 parameters for each one of them These parameters are divided as:

X, where X is a Data frame of the features values, each value being one feature, and being known values. Each column of X is an independent variable. Target is a vector of the target values, being the values, we will predict. Target has only one column and is the dependent/target variable.

X = df[['Potential', 'Crossing', 'Finishing', 'Heading Accuracy', 'Short Passing', 'Volleys', 'Dribbling', 'Curve', 'FKAccuracy', 'Long Passing', 'Ball Control', 'Acceleration', 'Sprint Speed', 'Agility', 'Reactions', 'Balance', 'Shot Power', 'Jumping', 'Stamina', 'Strength', 'Longshots', 'Aggression', 'Interceptions', 'Positioning', 'Vision', 'Penalties', 'Composure', 'Marking', 'Standing Tackle', 'Sliding Tackle', 'GK-Diving', 'GK-Handling', 'GK-Kicking', 'GK-Positioning', 'GK-Reflexes']]

3.2 Models

In this section, we will provide detailed descriptions of the different machine learning algorithms used in this model.

To build the Machine Learning Model which will predict the ratings of the football players and to make it a more accurate and highly efficient Machine Learning Model, the authors of this paper make use of the GridSearchCV cross – validation technique to build the model more accurate. Grid search cross-validation (GridSearchCV) is a powerful technique for hyperparameter tuning in machine learning. Figure 2. Depicts the correlation heatmap. The descriptive analysis of various Machine Learning Algorithms which are used to build this model are mentioned below:

1 Linear Regression-

- The authors of this paper chosen Linear regression as a primary algorithm in predicting the ratings of football players because of its simplicity and interpretability. It captures the fundamental essence of this works analysis and how these factors contribute to players performance by finding out the linear relationship between various parameters. Figure 3. Depicts the regression plot on Dribbling vs Ball Control with respect to the player's preferred foot.

2 Decision Tree Regressor-

- This algorithm offers a non-linear approach to rating prediction, consenting for the capture of complex relationship among player features. With its special ability to create hierarchical trees, the use of this algorithm can help us uncovering the nuanced dealings between various players ratings and attributes. Figure 4. Shows the importance feature in decision trees regressor.

3 Random Forest Regressor-

- This ML Algorithm extends upon the Decision Tree Regressor by leveraging ensemble learning techniques. Through combining the predictions from multiple trees, it improves the predictive accuracy and robustness, which makes this algorithm well-suited for identifying the uncertainty and the variability inherent in the player. This Algorithm excels in proper handling of the high dimensional data and mitigating overfitting. Figure 5. Depicts Importance features in Rando Forest Regressor.

4 Gradient Boosting Regressor-

- Gradient Boosting Regressor gives us an advanced ensemble learning technique that iteratively improves analytical models by merging the strengths of various weak learners,

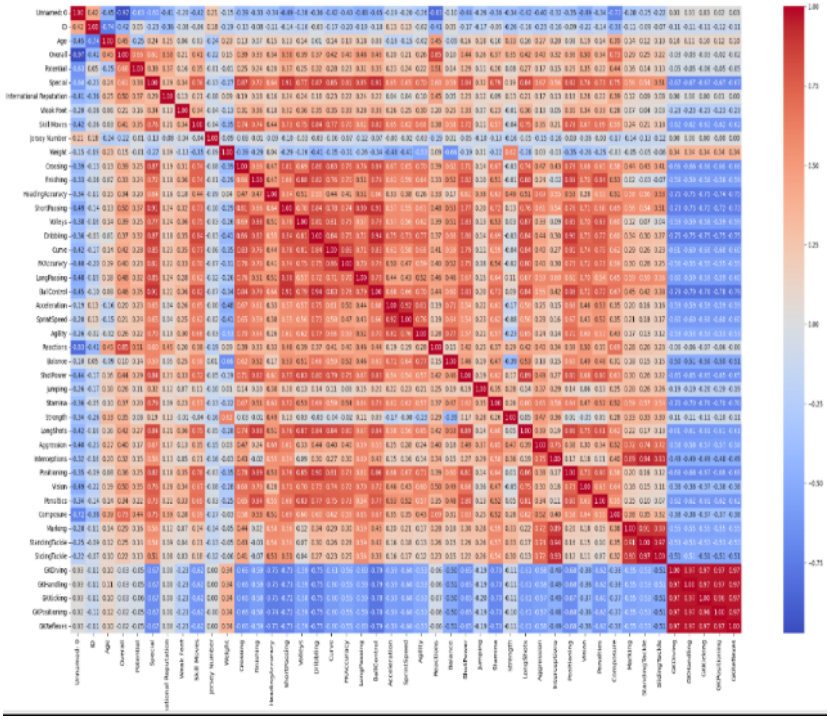


Figure 2: Correlation Heatmap

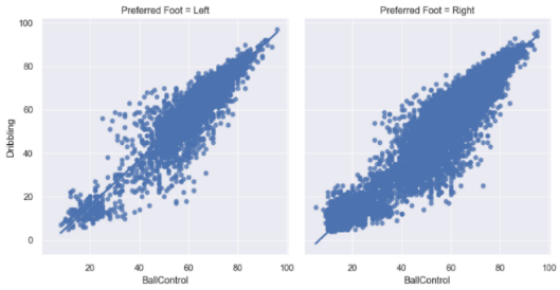


Figure 3: Dribbling v/s Ball Control

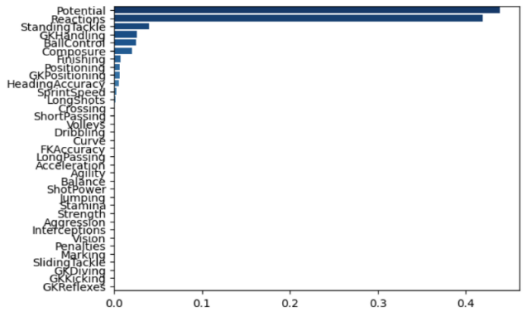


Figure 4: Importance feature in DTR

typically decision trees, etc. With the help of iterative error correction, Gradient Boosting Regressor captures the

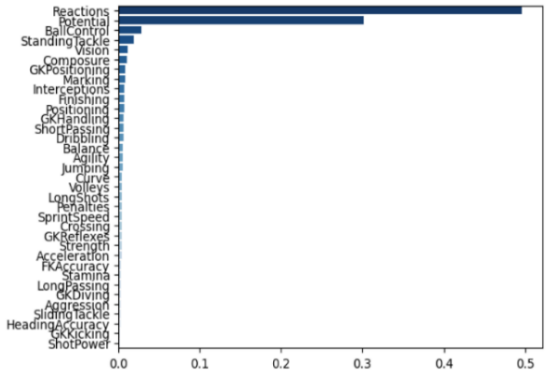


Figure 5: Importance feature in RFR

intricate patterns in the data, thus refining the overall predictive accuracy. With the help of this algorithm, we can particularly identify the subtle relationships and can handle the noisy data. Figure 6. Shows the importance feature in gradient boosting regressor.

5 Support Vector Regressor-

- SVR (Support Vector Regressor) positions out as a powerful algorithm designed explicitly for regression tasks. By plotting the input features into a high-dimensional space, it classifies hyperplanes due to which it best represents the

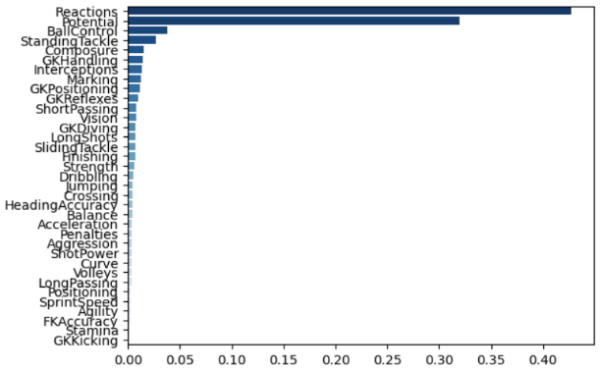


Figure 6: Importance feature in GBR

relations between the player attributes and ratings. This algorithm outshines the other in capturing the complex patterns and is

- Also robust to the outliers, making it suitable for modelling the football player ratings.

6 Voting Regressor-

- This Algorithm syndicates the predictions of various other individual regression models, by leveraging the miscellaneous algorithms or subgroups of data. By combining the outputs through weighted averages or majority voting, it connects the collective knowledge of different ML models to advance the overall predictive performance of a player.

7 Ridge Regression-

- This Regression Algorithm specifically addresses the multicollinearity among predictor variables, a very common challenge usually identified in rating prediction tasks or regression tasks. This Algorithm introduces regularization, which mitigates the overall impact of the correlated features, which further can be useful in improving the model's solidity and interpretability. This Algorithm is specifically used when we are dealing with the datasets having high correlation among the features such as ratings.

8 Lasso Regression-

- This Regression Algorithm serves as a regularization technique, particularly useful when dealing with high-dimensional datasets. By imposing sparsity through the feature selection, this ML Algorithm helps us in detecting the key attributes which contributes to player ratings while dropping the overfitting. Lasso Regression is well-matched for feature selection and regularization, making it an ideal algorithm when dealing in scenarios with many attributes.

9 Huber Regression-

- This Regression Algorithm offers robustness to outliers and noise in the data by balancing mean squared error and mean absolute error. This leads to compromise between the sensitivity of ordinary least squares regression and the robustness of absolute error-based techniques. Huber Regression is advantageous in scenarios where data quality varies and outliers are dominant, which guarantees reliable predictions as in the case of rating prediction of a football player.

10 K-Nearest Neighbours-

- K-Nearest Neighbours (KNN) Regression leverages the similarity of players based on their qualities to predict ratings. By averaging the ratings of k-nearest equals, it also offers an intuitive method to rating prediction, particularly when local affairs among qualities are important. KNN Regression is active in capturing neighbourhood effects and is well-suited for exploratory analysis.

11 Elastic-Net Regression-

- This Algorithm combines the strengths of the ridge and lasso regression models which makes it suitable for the datasets containing high collinearity among the predictor variables. This is done by striking a balance between regularization and feature selection, it addresses the challenges of multicollinearity while maintaining model interpretability.

4 RESULTS AND DISCUSSIONS

The objective is to predict the Football Players Ratings using machine learning. In this section, the authors of this paper discuss the results which the machine learning model is providing us with. Figure 7. Shows the OLS (Ordinary Least Squares) regression results Ordinary Least Squares (OLS) regression is a statistical method used to model the relationship between a dependent variable and one or more independent variables. Cross-validation techniques might also be employed to ensure the model's predictive accuracy. Table 1. displays the accuracy generated by various machine learning algorithms which are used in this model to predict the ratings of football players.

By observing the results generated through various pipelines in the model we can conclude that the Support Vector Regressor algorithm performs the best with 91.84% accuracy on the testing data followed by the Gradient Boosting Regressor with 90.78%, Voting Regressor with 91.68% and Random Forest Regressor with 88.89%. Apart from them the K-Nearest Neighbours Regression Algorithm highly overfits the model with 100% accuracy on the training set and 70.71% which shows the lack of feature engineering and improper hyperparameter tuning. The other performing algorithms in this project not seems to be performed very well with Ridge Regression giving the accuracy of 77.82%, Lasso Regression with 77.70%, Huber Regression with 74.30% and Elastic net Regression with 77.82% on the training data which shows their incapability to dealt with the linear parameters performing with the time. Meanwhile if we look at the Linear Regression Algorithm this seems to be performing pretty well as compared to others because it is showing the Mean Absolute percentage error of 1.51% on the training and 1.31% on the testing set which means it performs well with the linear parameters mentioned in the parameters and also with its popularity of being the most inconsistent in giving the best accuracy it performs really well.

Table 2. outlines the Ratings predicted by various ML Algorithms which are used to build this Machine Learning Model.

In summary, the use of algorithms such as Linear Regression, Decision Tree Regressor, Random Forest Regressor, Gradient Boosting Regressor, Voting Regressor, Ridge Regression, Lasso Regression,

OLS Regression Results

Dep. Variable:	Overall	R-squared (uncentered):	1.000
Model:	OLS	Adj. R-squared (uncentered):	1.000
Method:	Least Squares	F-statistic:	6.879e+04
Date:	Wed, 01 Nov 2023	Prob (F-statistic):	0.00
Time:	11:50:06	Log-Likelihood:	-1658.6
No. Observations:	900	AIC:	3387.
Df Residuals:	865	BIC:	3555.
Df Model:	35		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Potential	0.4641	0.017	27.192	0.000	0.431	0.498
Crossing	0.0178	0.008	2.299	0.022	0.003	0.033
Finishing	0.0049	0.009	0.534	0.593	-0.013	0.023
HeadingAccuracy	0.0237	0.007	3.236	0.001	0.009	0.038
ShortPassing	0.0271	0.016	1.686	0.092	-0.004	0.059
Volleys	0.0051	0.007	0.688	0.492	-0.009	0.020
Dribbling	0.0022	0.014	0.153	0.878	-0.026	0.030
Curve	-0.0066	0.008	-0.864	0.388	-0.022	0.008
FKAccuracy	0.0093	0.006	1.511	0.131	-0.003	0.021
LongPassing	0.0224	0.011	2.096	0.036	0.001	0.043
BallControl	0.0414	0.016	2.538	0.011	0.009	0.073
Acceleration	0.0074	0.013	0.565	0.572	-0.018	0.033
SprintSpeed	0.0072	0.011	0.635	0.526	-0.015	0.029
Agility	-0.0026	0.010	-0.256	0.798	-0.023	0.017
Reactions	0.2862	0.016	18.372	0.000	0.256	0.317
Balance	0.0134	0.008	1.713	0.087	-0.002	0.029
ShotPower	0.0109	0.009	1.197	0.232	-0.007	0.029
Jumping	0.0059	0.006	1.004	0.316	-0.006	0.017
Stamina	-0.0120	0.008	-1.497	0.135	-0.028	0.004
Strength	0.0366	0.008	4.685	0.000	0.021	0.052
LongShots	0.0082	0.009	0.936	0.350	-0.009	0.026
Aggression	0.0067	0.007	1.024	0.306	-0.006	0.019
Interceptions	0.0010	0.009	0.109	0.914	-0.018	0.020
Positioning	-0.0471	0.010	-4.752	0.000	-0.067	-0.028
Vision	-0.0202	0.009	-2.229	0.026	-0.038	-0.002
Penalties	0.0243	0.007	3.516	0.000	0.011	0.038
Composure	0.0690	0.011	6.307	0.000	0.048	0.090
Marking	0.0077	0.007	1.163	0.245	-0.005	0.021
StandingTackle	-0.0192	0.012	-1.656	0.098	-0.042	0.004
SlidingTackle	0.0107	0.010	1.045	0.296	-0.009	0.031
GKDiving	0.0409	0.016	2.585	0.010	0.010	0.072
GKHandling	0.0261	0.015	1.687	0.092	-0.004	0.056
GKKicking	-0.0103	0.014	-0.741	0.459	-0.038	0.017
GKPositioning	0.0327	0.016	2.082	0.038	0.002	0.063
GKReflexes	0.0333	0.016	2.113	0.035	0.002	0.064

Omnibus:	5.109	Durbin-Watson:	1.956
Prob(Omnibus):	0.078	Jarque-Bera (JB):	4.965
Skew:	-0.173	Prob(JB):	0.0835
Kurtosis:	3.112	Cond. No.	151.

Figure 7: OLS Regression Results

Table 1: Scores obtained through ML Algorithms

S No.	ML Algorithms	Training Data Accuracy (%)	Testing Data Accuracy (%)
1.	Linear Regression	1.51% (Mean Absolute Percentage Error)	1.31% (Mean Absolute Percentage Error)
2.	Decision Tree Regressor	83.97%	83.09%
3.	Random Forest Regressor	95.84%	88.89%
4.	Gradient Boosting Regressor	99.96%	90.78%
5.	Support Vector Regressor	93.83%	91.84%
6.	Voting Regressor	95.03%	91.68%
7.	Ridge Regression	77.82%	82.95%
8.	Lasso Regression	77.70%	82.94%
9.	Huber Regression	74.30%	78.55%
10.	K-Nearest Neighbours Regression	100.00%	70.71%
11.	Elastic-Net Regression	77.82%	82.96%

Table 2: Predicted Ratings

	ID	Rating	DTR	RFR	LR	GBR	SVR	VR	RR	LAR	HR	ER	KNR
0	185431	79	79.0	78.799917	77.849511	78.675771	78.633066	78.720731	78.508975	78.477000	77.882817	78.481253	78.593484
1	205633	78	78.0	77.944619	77.489967	78.269791	78.171388	78.080553	78.027854	78.042759	77.471586	78.039243	79.766013
2	171688	78	78.0	77.943286	78.496784	77.750785	78.898771	78.272957	78.796903	78.870193	78.494435	78.861601	79.358449
3	161648	79	78.0	78.921741	76.998893	78.728098	77.185848	78.036544	77.355345	77.349944	76.964402	77.347989	78.820661
4	231943	79	80.0	79.735038	82.361510	78.218492	80.073798	79.940327	81.668611	81.718385	82.436859	81.714525	79.454236
5	231478	79	79.0	79.523939	80.731546	78.515601	78.983187	79.292893	80.429378	80.436798	80.584450	80.433986	77.817851
6	213655	78	80.0	78.887655	78.612605	78.615793	77.373201	78.736751	78.803050	78.833622	78.534344	78.830285	79.073307
7	190149	79	79.0	78.754027	78.273538	78.647890	78.825926	78.774535	78.659020	78.772945	78.288803	78.763915	78.431405
8	49369	80	79.0	78.832591	77.306314	78.683452	79.268720	78.712630	77.795231	77.751254	77.352829	77.758818	78.744668
9	195861	80	80.0	79.742956	80.217062	80.307100	79.519587	79.938175	80.132103	80.041019	80.170552	80.051470	79.983675

[DTR=Decision Tree Regressor, RFR=Random Forest Regressor, LR=Linear Regression, GBR=Gradient Boosting Regressor, SVR=Support Vector Regressor, VR=Voting Regressor, RR=Ridge Regression, LAR=Lasso Regression, HR=Huber Regression, ER=Elastic-Net Regression, KNR=K-Nearest Neighbours Regression]

Huber Regression, K-Nearest Neighbours Regression and Elastic-Net Regression in combination can result in improved prediction accuracy, robustness, and the ability to handle complex relationships between features and the target variable. This ensemble approach contributes to the creation of a more reliable and versatile model for the prediction of Ratings of the Football players.

5 CONCLUSION

In this study, the researchers investigated the various machine learning algorithms based on the accuracy they generate for the prediction of football players ratings. The methodologies used in the range of models including linear regression, decision tree regressor, random forest regressor, gradient boosting regressor, support vector regressor, voting regressor, ridge regression, lasso regression, k-nearest neighbour's regression, Huber regression, and elastic-net regression. Through a comprehensive and rigorous analysis, it is concluded that each of the ML Algorithms exhibited unique strengths in capturing the complex relationships within the dataset. Notably, ensemble methods such as Support Vector Regressor and Gradient Boosting showcased superior predictive accuracy, while linear regression is good enough to compete with them. Furthermore, regularization techniques like Ridge and Lasso regression contributed to robustness and interpretability in the predictive models. The conclusions drawn underscore the critical importance of judiciously selecting algorithms tailored to the specific characteristics of the dataset for precise and reliable player rating predictions.

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